MERN Journal Practicals

Code in Practical Order

Copy-friendly reference for VS Code — JavaScript, React, Express and MongoDB

Important note: This PDF compiles the reconstructed code files available in the code pack. The uploaded journal is a 55-page scanned document, and several handwritten pages are not reliably readable. This is not guaranteed to be a word-for-word transcription of every journal practical. Missing entries are listed at the end rather than being falsely presented as verified.

How to use: Each section shows the practical folder and filename. For Node.js files, open the terminal in that folder and run `node app.js` (or `node server.js`). React files belong inside a Vite React project; follow the corresponding README where provided.

Start Here

```
# MERN Journal Practicals — Code in Order

Each practical is in its own folder. Open the relevant folder in VS Code and follow the README inside it. Plain JavaScript

This pack reconstructs the code from the uploaded scanned journal and the practicals confirmed in chat. Some handwriting/p
```

01A-node-version

01A-node-version/README.md

````
# 1A — Verify Node.js and npm
No source file needed. In VS Code terminal run:
```powershell
node --version
npm --version
```
````

01B-console-output

01B-console-output/README.md

````
# Run instructions

Open this folder in VS Code Terminal and run:

```powershell
node app.js
```

For Express folders, first run `npm init -y` and `npm install express`, then `node server.js`. For MongoDB folders, run `n
````

01B-console-output/app.js

```
console.log("Hello World");
console.log("Welcome to Node.js");
```

01C-student-details

01C-student-details/README.md

```
# Run instructions

Open this folder in VS Code Terminal and run:

```powershell
node app.js
```

For Express folders, first run `npm init -y` and `npm install express`, then `node server.js`. For MongoDB folders, run `n
```

01C-student-details/app.js

```
let name = "Eren";
let age = 20;
let course = "BScIT";

console.log("Student Name:", name);
console.log("Age:", age);
console.log("Course:", course);
```

01D-functions

01D-functions/README.md

````
# Run instructions

Open this folder in VS Code Terminal and run:

```powershell
node app.js
```

For Express folders, first run `npm init -y` and `npm install express`, then `node server.js`. For MongoDB folders, run `n
````

01D-functions/app.js

```
function display(name, age) {
  console.log("Student Name:", name);
  console.log("Student Age:", age);
}

display("Eren", 20);
```

01E-addition-function

01E-addition-function/README.md

````
# Run instructions

Open this folder in VS Code Terminal and run:

```powershell
node app.js
```

For Express folders, first run `npm init -y` and `npm install express`, then `node server.js`. For MongoDB folders, run `n
````

01E-addition-function/app.js

```
function add(a, b) {
  return a + b;
}

console.log("Addition:", add(10, 20));
console.log("Addition:", add(72, 63));
```

01F-object-basics

01F-object-basics/README.md

````
# Run instructions

Open this folder in VS Code Terminal and run:

```powershell
node app.js
```

For Express folders, first run `npm init -y` and `npm install express`, then `node server.js`. For MongoDB folders, run `n
````

01F-object-basics/app.js

```
let student = {
  name: "Eren",
  age: 20,
  course: "BScIT"
};

console.log("Name:", student.name);
console.log("Age:", student.age);
console.log("Course:", student.course);
```

02A-arithmetic

02A-arithmetic/README.md

````
# Run instructions

Open this folder in VS Code Terminal and run:

```powershell
node app.js
```

For Express folders, first run `npm init -y` and `npm install express`, then `node server.js`. For MongoDB folders, run `n
````

02A-arithmetic/app.js

```
let a = 10;
let b = 5;
console.log("Addition:", a + b);
console.log("Subtraction:", a - b);
console.log("Multiplication:", a * b);
console.log("Division:", a / b);
```

02B-conditionals

02B-conditionals/README.md

````
# Run instructions

Open this folder in VS Code Terminal and run:

```powershell
node app.js
```

For Express folders, first run `npm init -y` and `npm install express`, then `node server.js`. For MongoDB folders, run `n
````

02B-conditionals/app.js

```
let marks = 75;
if (marks >= 40) {
  console.log("Pass");
} else {
  console.log("Fail");
}
```

02C-array-basics

02C-array-basics/README.md

````
# Run instructions

Open this folder in VS Code Terminal and run:

```powershell
node app.js
```

For Express folders, first run `npm init -y` and `npm install express`, then `node server.js`. For MongoDB folders, run `n
````

02C-array-basics/app.js

```
let numbers = [10, 20, 30, 40, 50];
console.log("Array:", numbers);
console.log("First element:", numbers[0]);
console.log("Array length:", numbers.length);
```

02D-array-map-filter-reduce

02D-array-map-filter-reduce/README.md

````
# Run instructions

Open this folder in VS Code Terminal and run:

```powershell
node app.js
```

For Express folders, first run `npm init -y` and `npm install express`, then `node server.js`. For MongoDB folders, run `n
````

02D-array-map-filter-reduce/app.js

```
let numbers = [10, 20, 30, 40, 50];
let filtered = numbers.filter(num => num > 25);
let doubled = numbers.map(num => num * 2);
let sum = numbers.reduce((total, num) => total + num, 0);

console.log("Original Array:", numbers);
console.log("Double Array:", doubled);
console.log("Filtered Array:", filtered);
console.log("Sum of array:", sum);
```

02E-object-manipulation

02E-object-manipulation/README.md

````
# Run instructions

Open this folder in VS Code Terminal and run:

```powershell
node app.js
```

For Express folders, first run `npm init -y` and `npm install express`, then `node server.js`. For MongoDB folders, run `n
````

02E-object-manipulation/app.js

```
let student = { name: "Eren", age: 20, course: "BScIT" };
console.log("Original student details:");
console.log(student);

student.age = 21;
student.city = "Mumbai";
delete student.course;

console.log("\nUpdated student details:");
console.log(student);
```

02F-student-function

02F-student-function/README.md

````
# Run instructions

Open this folder in VS Code Terminal and run:

```powershell
node app.js
```

For Express folders, first run `npm init -y` and `npm install express`, then `node server.js`. For MongoDB folders, run `n
````

02F-student-function/app.js

```
function displayStudent(student) {
  console.log("Student Name:", student.name);
  console.log("Roll No:", student.rollNo);
  console.log("Marks:", student.marks);
}

displayStudent({ name: "Eren", rollNo: 101, marks: 84 });
```

03A-react-basic

03A-react-basic/README.md

```
# 3A — Basic React app
Create the project with `npm create vite@latest mern-practical -- --template react`, then `cd mern-practical`, `npm insta
```

03A-react-basic/src-App.jsx

```
function App() {
  return (
    <div>
      <h1>Student Information</h1>
      <p>Name: Eren</p>
      <p>Course: BScIT</p>
      <p>College: My College</p>
    </div>
  );
}
export default App;
```

03B-react-components

03B-react-components/src-App.jsx

```
function Header() { return <h1>Welcome to React</h1>; }
function Content() { return <p>This is the content component.</p>; }
function Footer() { return <footer>Copyright 2026</footer>; }

function App() {
  return <div><Header /><Content /><Footer /></div>;
}
export default App;
```

03C-react-props

03C-react-props/src-App.jsx

```
function Student(props) {
  return <div><h2>{props.name}</h2><p>Course: {props.course}</p></div>;
}
function App() {
  return <div><h1>Student Details</h1><Student name="Eren" course="BScIT" /></div>;
}
export default App;
```

03D-react-list

03D-react-list/src-App.jsx

```
function App() {
  const students = ["Eren", "Mikasa", "Armin"];
  return <div><h1>Student List</h1><ul>{students.map((name, index) => <li key={index}>{name}</li>)}</ul></div>;
}
export default App;
```

04A-useState-counter

04A-useState-counter/src-App.jsx

```
import { useState } from "react";
function App() {
  const [count, setCount] = useState(0);
  return <div><h1>Counter: {count}</h1><button onClick={() => setCount(count + 1)}>Increment</button><button onClick={() =
}
export default App;
```

04B-react-form

04B-react-form/src-App.jsx

```
import { useState } from "react";
function App() {
  const [name, setName] = useState("");
  const [submitted, setSubmitted] = useState("");
  function handleSubmit(event) { event.preventDefault(); setSubmitted(name); }
  return <div><h1>Student Form</h1><form onSubmit={handleSubmit}><input value={name} onChange={e => setName(e.target.value
}
export default App;
```

04C-react-conditional

04C-react-conditional/src-App.jsx

```
import { useState } from "react";
function App() {
  const [show, setShow] = useState(true);
  return <div><button onClick={() => setShow(!show)}>{show ? "Hide" : "Show"} Message</button>{show && <p>Welcome to React
}
export default App;
```

05A-react-router

05A-react-router/README.md

```
# 5A — React Router starter
Inside a Vite React project, run `npm install react-router-dom`. Put `src/main.jsx` and `src/App.jsx` from this folder int
```

05A-react-router/src-App.jsx

```
import { Link, Routes, Route } from "react-router-dom";
function Home() { return <h2>Home Page</h2>; }
function About() { return <h2>About Page</h2>; }
function App() { return <div><h1>React Router Example</h1><nav><Link to="/">Home</Link> | <Link to="/about">About</Link><A
export default App;
```

05A-react-router/src-main.jsx

```
import React from "react";
import ReactDOM from "react-dom/client";
import { BrowserRouter } from "react-router-dom";
import App from "./App.jsx";
import "./index.css";
ReactDOM.createRoot(document.getElementById("root")).render(<React.StrictMode><BrowserRouter><App /></BrowserRouter></Reac
```

05B-react-router-users

05B-react-router-users/README.md

```
# 5B — Users route
Install `npm install react-router-dom`. Use `src-main.jsx` as `src/main.jsx`, and `src-App.jsx` as `src/App.jsx` in a Vite
```

05B-react-router-users/src-App.jsx

```
import { Link, Routes, Route } from "react-router-dom";
function Users() { return <div><h2>Users Page</h2><p>This page will display data from an API.</p></div>; }
function App() { return <div><h1>React Route Example</h1><Link to="/users">Users</Link><Routes><Route path="/users" eleme
export default App;
```

05B-react-router-users/src-main.jsx

```
import React from "react";
import ReactDOM from "react-dom/client";
import { BrowserRouter } from "react-router-dom";
import App from "./App.jsx";
ReactDOM.createRoot(document.getElementById("root")).render(<BrowserRouter><App /></BrowserRouter>);
```

05D-fetch-users-axios

05D-fetch-users-axios/README.md

```
# 5D — Fetch users with Axios
In Vite React project run `npm install axios react-router-dom`. Use `src-main.jsx` as `src/main.jsx`, `src-App.jsx` as `sr
```

05D-fetch-users-axios/Users.jsx

```
import { useEffect, useState } from "react";
import axios from "axios";
export default function Users() {
  const [users, setUsers] = useState([]);
  useEffect(() => { axios.get("https://jsonplaceholder.typicode.com/users").then(res => setUsers(res.data)).catch(err => c
  return <div><h2>Users List</h2><ul>{users.map(user => <li key={user.id}>{user.name}</li>)}</ul></div>;
}
```

05D-fetch-users-axios/src-App.jsx

```
import { Link, Routes, Route } from "react-router-dom";
import Users from "./pages/Users.jsx";
function App() { return <div><h1>React Route Example</h1><nav><Link to="/users">Users</Link></nav><Routes><Route path="/us
export default App;
```

05D-fetch-users-axios/src-main.jsx

```
import React from "react"; import ReactDOM from "react-dom/client"; import { BrowserRouter } from "react-router-dom"; impo
ReactDOM.createRoot(document.getElementById("root")).render(<React.StrictMode><BrowserRouter><App /></BrowserRouter></Reac
```

05E-users-table

05E-users-table/Users.jsx

```
import { useEffect, useState } from "react";
import axios from "axios";
export default function Users() {
  const [users, setUsers] = useState([]);
  useEffect(() => { axios.get("https://jsonplaceholder.typicode.com/users").then(res => setUsers(res.data)).catch(err => c
  return <div><h2>Users Table</h2><table border="1" cellPadding="8"><thead><tr><th>ID</th><th>Name</th><th>Email</th></tr>
}
```

05F-search-users

05F-search-users/Users.jsx

```
import { useEffect, useState } from "react";
import axios from "axios";
export default function Users() {
  const [users, setUsers] = useState([]); const [search, setSearch] = useState("");
  useEffect(() => { axios.get("https://jsonplaceholder.typicode.com/users").then(res => setUsers(res.data)).catch(err => c
  const filteredUsers = users.filter(user => user.name.toLowerCase().includes(search.toLowerCase()));
  return <div><h2>Search Users</h2><input placeholder="Search user" value={search} onChange={e => setSearch(e.target.value
}
```

06A-express-server

06A-express-server/README.md

````
# Run instructions

Open this folder in VS Code Terminal and run:

```powershell
node app.js
```

For Express folders, first run `npm init -y` and `npm install express`, then `node server.js`. For MongoDB folders, run `n
````

06A-express-server/server.js

```
const express = require("express");
const app = express();
const PORT = 3000;
app.get("/", (req, res) => res.send("Welcome to Express Server"));
app.listen(PORT, () => console.log(`Server running at http://localhost:${PORT}`));
```

06B-express-routes

06B-express-routes/README.md

```
# Run instructions

Open this folder in VS Code Terminal and run:

```powershell
node app.js
```

For Express folders, first run `npm init -y` and `npm install express`, then `node server.js`. For MongoDB folders, run `n
```

06B-express-routes/server.js

```
const express = require("express");
const app = express(); const PORT = 3000;
app.get("/", (req, res) => res.send("Home Page"));
app.get("/about", (req, res) => res.send("About Page"));
app.get("/contact", (req, res) => res.send("Contact Page"));
app.listen(PORT, () => console.log(`Server running at http://localhost:${PORT}`));
```

06C-express-json-api

06C-express-json-api/README.md

````
# Run instructions

Open this folder in VS Code Terminal and run:

```powershell
node app.js
```

For Express folders, first run `npm init -y` and `npm install express`, then `node server.js`. For MongoDB folders, run `n
````

06C-express-json-api/server.js

```
const express = require("express");
const app = express(); const PORT = 3000;
app.use(express.json());
app.get("/api", (req, res) => res.json({ message: "Welcome to API" }));
app.get("/api/student", (req, res) => res.json({ name: "Eren", rollNo: 101, course: "TYIT" }));
app.listen(PORT, () => console.log(`Server running at http://localhost:${PORT}`));
```

06D-express-query-params

06D-express-query-params/README.md

```
# Run instructions

Open this folder in VS Code Terminal and run:

```powershell
node app.js
```

For Express folders, first run `npm init -y` and `npm install express`, then `node server.js`. For MongoDB folders, run `n
```

06D-express-query-params/server.js

```
const express = require("express");
const app = express(); const PORT = 3000;
app.get("/user", (req, res) => { const name = req.query.name || "Guest"; res.send(`Hello, ${name}`); });
app.listen(PORT, () => console.log(`Server running at http://localhost:${PORT}`));
```

06E-express-multiple-routes

06E-express-multiple-routes/README.md

````
# Run instructions

Open this folder in VS Code Terminal and run:

```powershell
node app.js
```

For Express folders, first run `npm init -y` and `npm install express`, then `node server.js`. For MongoDB folders, run `n
````

06E-express-multiple-routes/server.js

```
const express = require("express");
const app = express(); const PORT = 3000;
app.get("/", (req, res) => res.send("Welcome to Express Server"));
app.get("/api", (req, res) => res.json({ message: "Welcome to API" }));
app.get("/home", (req, res) => res.send("Home Page"));
app.listen(PORT, () => console.log(`Server running at http://localhost:${PORT}`));
```

06F-express-post

06F-express-post/README.md

```
# Run instructions

Open this folder in VS Code Terminal and run:

```powershell
node app.js
```

For Express folders, first run `npm init -y` and `npm install express`, then `node server.js`. For MongoDB folders, run `n
```

06F-express-post/server.js

```
const express = require("express");
const app = express(); const PORT = 3000;
app.use(express.json());
app.post("/student", (req, res) => res.status(201).json({ message: "Student received", student: req.body }));
app.listen(PORT, () => console.log(`Server running at http://localhost:${PORT}`));
```

07A-mongodb-connect

07A-mongodb-connect/README.md

```
# Run instructions

Open this folder in VS Code Terminal and run:

```powershell
node app.js
```

For Express folders, first run `npm init -y` and `npm install express`, then `node server.js`. For MongoDB folders, run `n
```

07A-mongodb-connect/app.js

```
const { MongoClient } = require("mongodb");
const client = new MongoClient("mongodb://127.0.0.1:27017");
async function main() { try { await client.connect(); console.log("Connected to MongoDB"); const db = client.db("college")
main();
```

07B-mongodb-insert

07B-mongodb-insert/README.md

```
# Run instructions

Open this folder in VS Code Terminal and run:

```powershell
node app.js
```

For Express folders, first run `npm init -y` and `npm install express`, then `node server.js`. For MongoDB folders, run `n
```

07B-mongodb-insert/app.js

```
const { MongoClient } = require("mongodb");
const client = new MongoClient("mongodb://127.0.0.1:27017");
async function main() { try { await client.connect(); const collection = client.db("college").collection("students"); cons
main();
```

07C-mongodb-read-update-delete

07C-mongodb-read-update-delete/README.md

````
# Run instructions

Open this folder in VS Code Terminal and run:

```powershell
node app.js
```

For Express folders, first run `npm init -y` and `npm install express`, then `node server.js`. For MongoDB folders, run `n
````

07C-mongodb-read-update-delete/app.js

```
const { MongoClient, ObjectId } = require("mongodb");
const client = new MongoClient("mongodb://127.0.0.1:27017");
async function main() { try {
  await client.connect(); const collection = client.db("college").collection("students");
  const inserted = await collection.insertOne({ name: "Armin", rollNo: 102, course: "TYIT" });
  console.log("Inserted:", inserted.insertedId.toString());
  console.log("All students:", await collection.find({}).toArray());
  await collection.updateOne({ _id: inserted.insertedId }, { $set: { course: "MERN" } });
  console.log("Updated student:", await collection.findOne({ _id: inserted.insertedId }));
  await collection.deleteOne({ _id: inserted.insertedId }); console.log("Student deleted");
} catch (err) { console.error(err); } finally { await client.close(); } }
main();
```

Practical coverage notes

Included code folders in this PDF:

01A-node-version, 01B-console-output, 01C-student-details, 01D-functions, 01E-addition-function, 01F-object-basics, 02A-arithmetic, 02B-conditionals, 02C-array-basics, 02D-array-map-filter-reduce, 02E-object-manipulation, 02F-student-function, 03A-react-basic, 03B-react-components, 03C-react-props, 03D-react-list, 04A-useState-counter, 04B-react-form, 04C-react-conditional, 05A-react-router, 05B-react-router-users, 05D-fetch-users-axios, 05E-users-table, 05F-search-users, 06A-express-server, 06B-express-routes, 06C-express-json-api, 06D-express-query-params, 06E-express-multiple-routes, 06F-express-post, 07A-mongodb-connect, 07B-mongodb-insert, 07C-mongodb-read-update-delete

The current reconstructed pack does not contain verified source files for Practical 5C or Practical 8A–8C. Some practical labels and exact wording could not be confidently read from the handwritten scan. Please use this PDF as the available code compilation, not as a claim that every original journal page has been transcribed exactly.